

Assignment: 14

Problem Statement: Create an Elastic IP for an Instance.

Step 1: Create an EC2 instance as we've created before.

aws

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SICKY KUMAR (7225-6022-4322)

SICKY KUMAR

EC2 > Instances > Launch an instance

Name and tags

Name

newinst

Add additional tags

▼ Application and OS Images (Amazon Machine Image)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

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▼ Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.7.2...read more

ami-0f1dcc636b69a6438

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

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EC2 > Instances > Launch an instance

Key pair name - required

pr-51

Create new key pair

▼ Network settings

Network

vpc-0e2181ce3777a313c

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups

Select security groups

secure51 sg-07c04a05d7396992f

Compare security group rules

Edit

▼ Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0e35ddab05955cf57

Virtual server type (instance type)

t2.micro

Firewall (security group)

secure51

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

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EC2 > Instances > Launch an instance

User data - optional

Upload a file with your user data or enter it in the field.

Choose file

```
#!/bin/bash
apt-get update
apt-get upgrade
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/PrasunK8/Prasun_aws.git
cd Prasun_aws
npm install
node index.js
```

☐ User data has already been base64 encoded

▼ Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0e35ddab05955cf57

Virtual server type (instance type)

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Firewall (security group)

secure51

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1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

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EC2 > Instances > Launch an instance

Success
Successfully initiated launch of instance (i-0cf4255a187c8ad14)

Step 2: Go to instance and Restart (stop and then start again) the created instance

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EC2 > Instances

Instances (1/3) Info Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

☐ inst1 i-01ae128cda6e8c8e0 Terminated [View alarms](#) ap-s

☐ inst51 i-0f6971ea8b7ea7caa Terminated [View alarms](#) ap-s

☒ newinst i-0cf4255a187c8ad14 Running [View alarms](#) ap-s

t2.micro Initializing

[Stop instance](#)
[Start instance](#)
[Reboot instance](#)
[Hibernate instance](#)
[Terminate \(delete\) instance](#)

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EC2 > Instances

Instances (1/3) Info Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

☒ newinst i-0cf4255a187c8ad14 Stopped [View alarms](#) ap-s

inst1 i-01ae128cda6e8c8e0 Terminated [View alarms](#) ap-s

inst51 i-0f6971ea8b7ea7caa Terminated [View alarms](#) ap-s

t2.micro -

[Stop instance](#)
[Start instance](#)
[Reboot instance](#)
[Hibernate instance](#)
[Terminate \(delete\) instance](#)

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EC2 > Instances > i-0cf4255a187c8ad14

Instance summary for i-0cf4255a187c8ad14 (newinst) Info [Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID
i-0cf4255a187c8ad14

IPv6 address
-

Public IPv4 address
3.111.35.214 [open address](#)

Instance state
Running

Private IPv4 addresses
172.31.10.116

Public IPv4 DNS
ec2-3-111-35-214.ap-south-1.compute.amazonaws.com [open address](#)

Hostname type
IP name: ip-172-31-10-116.ap-south-1.compute.internal

Private IP DNS name (IPv4 only)
ip-172-31-10-116.ap-south-1.compute.internal

Instance type
t2.micro

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
3.111.35.214 [Public IP]

VPC ID
vpc-0e2181ce3777a313c

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. [Learn more](#)

Step 3: Go to Elastic IPs and click on Allocate IP to create an Elastic IP and Associate it.

This screenshot shows the AWS console's 'Elastic IP addresses' page. The left sidebar contains the 'Elastic Block Store' and 'Network & Security' sections. The main content area has a search bar and a table header with columns: Name, Allocated IPv4 address, Type, Allocation ID, and Revers. Below the header, a message states 'No Elastic IP addresses found in this Region'.

This screenshot shows the header of the 'Allocate Elastic IP address' page. The breadcrumb trail is 'EC2 > Elastic IP addresses > Allocate Elastic IP address'. The page title is 'Allocate Elastic IP address' with an 'Info' link.

This screenshot shows the 'Elastic IP address settings' section. It includes a 'Public IPv4 address pool' with three radio button options: 'Amazon's pool of IPv4 addresses' (selected), 'Public IPv4 address that you bring to your AWS account with BYOIP', and 'Customer-owned pool of IPv4 addresses'. Below this is a 'Network border group' search box containing 'ap-south-1'.

This screenshot shows the AWS console after a successful Elastic IP allocation. A green banner at the top states 'Elastic IP address allocated successfully. Elastic IP address 3.109.136.158' with an 'Associate this Elastic IP address' button. Below, the 'Elastic IP addresses (1)' table shows one entry: a Public IP with address 3.109.136.158, Allocation ID eipalloc-0680d9481cc977267, and Type Public IP.

This screenshot shows the details of the allocated Elastic IP address 3.109.136.158. The 'Summary' section contains the following information:

Field	Value
Allocated IPv4 address	3.109.136.158
Type	Public IP
Allocation ID	eipalloc-0680d9481cc977267
Reverse DNS record	-
Association ID	-
Scope	VPC
Associated instance ID	-
Private IP address	-
Network interface ID	-
Network interface owner account ID	-
Public DNS	-
NAT Gateway ID	-
Address pool	Amazon
Network border group	ap-south-1

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EC2 > Elastic IP addresses > 3.109.136.158 > Associate Elastic IP address

Associate Elastic IP address

Choose the instance or network interface to associate to this Elastic IP address (3.109.136.158)

Elastic IP address: 3.109.136.158

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance

☐ Network interface

⚠ If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance

Private IP address
The private IP address with which to associate the Elastic IP address.

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

Step 4: Check the Instance IP has been changed as same as the newly created Elastic IP.

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EC2 > Instances > i-0cf4255a187c8ad14

EC2

Dashboard

EC2 Global View

Events

▼ **Instances**

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

▼ **Images**

AMIs

AMI Categories

Instance summary for i-0cf4255a187c8ad14 (newinst) [Info](#) [Connect](#) [Instance state](#) [Actions](#)

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Instance ID i-0cf4255a187c8ad14 IPv6 address — Hostname type IP name: ip-172-31-10-116.ap-south-1.compute.internal Answer private resource DNS name IPv4 (A) Auto-assigned IP address —	Public IPv4 address 3.109.136.158 open address Instance state Running Private IP DNS name (IPv4 only) ip-172-31-10-116.ap-south-1.compute.internal Instance type t2.micro VPC ID vpc-0e2181ce3777a313c	Private IPv4 addresses 172.31.10.116 Public IPv4 DNS ec2-3-109-136-158.ap-south-1.compute.amazonaws.com open address Elastic IP addresses 3.109.136.158 [Public IP] AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations
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So, we've successfully Created an Elastic IP for an instance.

At last Delete(Disassociate > Release) the Elastic IP and the created instance.